ISAC Model for RIS-Assisted Secure UAV Communication and Sensing

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Abstract

This report presents the implemented integrated sensing and communication (ISAC)-based UAV framework for secure wireless communication and target sensing. The proposed architecture consists of a multi-antenna base station (BS), a passive reconfigurable intelligent surface (RIS)-mounted UAV, a full-duplex cooperative jammer UAV, a mobile user, a vehicle sensing target, and homogeneous Poisson point process (HPPP)-distributed eavesdroppers. The implemented framework integrates RIS-assisted secure communication, cooperative jamming, vehicle reflection modeling, bistatic sensing, Fisher Information Matrix (FIM)-based Cramér–Rao Bound (CRB) analysis, energy-detection-based target detection probability, and a joint secrecy–sensing optimization framework. Furthermore, a four-agent multi-agent deep reinforcement learning (MADRL) framework employing MAPPO, MATD3PG, and MADDPG algorithms, together with a Random Walk mobility baseline, is implemented to jointly optimize BS beamforming, UAV trajectory, jammer beamforming, and RIS phase configuration under dynamic network conditions. Experimental results demonstrate the convergence characteristics of the proposed learning algorithms and validate the implemented communication and sensing modules through secrecy rate, CRB, and detection probability performance metrics. The report summarizes the mathematical formulation, implementation methodology, experimental evaluation, and current scope and limitations of the developed ISAC framework.

Index Terms

Integrated sensing and communication, UAV communications, reconfigurable intelligent surface, physical layer security, Cramer–Rao bound, detection probability, multi-agent reinforcement learning.

I. Introduction

INTEGRATED sensing and communication (ISAC) systems aim to reuse a single wireless infrastructure for simultaneous data transmission and environmental awareness, thereby improving spectral, hardware, and energy efficiency relative to separately operated radar and communication systems. Unmanned aerial vehicle (UAV) platforms are particularly attractive for ISAC deployment because they offer controllable three-dimensional geometry, predominantly line-of-sight (LoS) air-to-ground propagation, and rapid, on-demand repositioning. However, the same favorable LoS propagation that benefits legitimate users also exposes confidential transmissions to interception by passive eavesdroppers, motivating the joint treatment of physical-layer security and sensing within a single UAV-assisted ISAC architecture.

To address this joint communication–security–sensing problem, the implemented system combines a broad set of complementary techniques. On the communication side, a passive reconfigurable intelligent surface (RIS)-assisted UAV establishes a controllable reflection path between the base station (BS) and the ground user through a diagonal, unit-modulus phase-shift matrix, compensating for the absence of a reliable direct BS-to-user link, while a second, dedicated UAV performs cooperative full-duplex jamming by transmitting artificial interference toward spatially distributed eavesdroppers so as to degrade their achievable rate while the legitimate link is preserved through directional beamforming, thereby enlarging the secrecy rate. The eavesdropper population itself is not treated as fixed but is instead modeled through stochastic geometry, with eavesdropper locations drawn from a homogeneous Poisson point process (HPPP), which allows secrecy performance to be evaluated over random, spatially unknown adversary deployments rather than a single deterministic layout. All of the underlying wireless links – BS–RIS, RIS–user, RIS–eavesdropper, jammer–user, and jammer–eavesdropper – are modeled using a combination of large-scale distance-dependent path loss and small-scale Rician fading, capturing the mixed line-of-sight/non-line-of-sight nature typical of aerial channels.

On the sensing side, ground vehicle targets are represented as point reflectors whose reflected signal strength depends on an aspect-angle-dependent radar cross section (RCS), which provides a physically motivated basis for generating the sensing channel. The sensing platform itself is modeled as a uniform linear array (ULA), and steering-vector-based target response matrices are used to construct the composite sensing channel needed for angle-domain estimation. The achievable target-angle estimation accuracy is then quantified through the Fisher information matrix (FIM) and its inverse, the Cramer–Rao bound (CRB), which lower-bounds the variance of any unbiased angle estimator and serves as the sensing-accuracy metric inside the optimization and learning objectives described later. Target presence, in turn, is formulated as a binary hypothesis test and evaluated using an energy detector and a generalized-likelihood-ratio-test (GLRT)-style statistic, characterized through the detection probability P_D, the false-alarm probability P_{FA}, and receiver operating characteristic (ROC) curves. Because the

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monostatic sensing return – obtained with co-located transmit and receive elements at the sensing UAV – and the bistatic sensing return – obtained with a separated illuminator and receiver – generally experience independent fading, a selection-combining (SC) rule is additionally applied across the two sensing paths, retaining whichever branch yields the higher instantaneous sensing SNR at any given time.

Finally, these communication and sensing objectives are drawn together through a joint secrecy–sensing optimization framework, in which a weighted scalar objective combines the secrecy rate and the CRB-derived sensing utility under power, RIS, and mobility feasibility constraints, enabling an explicit secrecy–sensing trade-off analysis. Because this joint problem is non-convex and difficult to solve in closed form, it is instead addressed through multi-agent deep reinforcement learning (MADRL), in which cooperative agents controlling BS beamforming, RIS-UAV trajectory, RIS phase configuration, and jammer beamforming are trained using multi-agent proximal policy optimization (MAPPO) and multi-agent twin-delayed deep deterministic policy gradient (MATD3PG)-style algorithms to learn control policies that maximize the joint secrecy–sensing reward.

The implemented network architecture is built around two cooperating UAVs with distinct and complementary roles: a passive *RIS-mounted UAV*, which relays and reflects the BS-to-user communication signal along a controllable phase-shifted path, and a *full-duplex jammer UAV*, which broadcasts artificial interference to suppress eavesdropper reception while additionally participating in the sensing pipeline. Together with a multi-antenna BS, a mobile ground user, a radar-relevant vehicle target, and an HPPP field of eavesdroppers, these two UAVs form the complete ISAC network whose geometry, communication model, and sensing model are formalized in Section II. The system evaluates three principal performance metrics: the secrecy rate R_s, the CRB-based sensing accuracy, and the detection probability P_D, and the remainder of this report describes the mathematical model, the implemented methodology, representative validation results, and the current scope and limitations of the ISAC implementation.

II. System Model and Problem Formulation

A. Network Geometry

The implemented ISAC network consists of six categories of nodes, each with a distinct functional role, as illustrated in Fig. 1:

- **Base station (BS):** An M_B-antenna ground BS located at $\mathbf{q}_B \in \mathbb{R}^3$. The BS is the sole source of the information-bearing signal and transmits a beamformed communication signal $x_B[n]$ using beamforming vector $\mathbf{w}_B[n] \in \mathbb{C}^{M_B \times 1}$, subject to a maximum transmit power constraint. There is no reliable direct link from the BS to the user; the BS relies on the RIS-mounted UAV to reach the user.
- **RIS-mounted UAV (relay/reflector UAV):** A UAV located at $\mathbf{q}_R[n] \in \mathbb{R}^3$, carrying a passive reconfigurable intelligent surface with N_R reflecting elements. Its role is purely one of *communication assistance*: it applies a diagonal, unit-modulus phase-shift matrix $\mathbf{\Phi}[n]$ to the incident BS signal and re-radiates it toward the user, forming the only usable BS-to-user propagation path. This UAV is also treated as the primary geometric sensing platform: its own array is used as a co-located (monostatic) transmit/receive array for target sensing.
- **Full-duplex jammer UAV (cooperative-jamming UAV):** A second UAV located at $\mathbf{q}_J[n] \in \mathbb{R}^3$, equipped with N_J antennas and operating in full-duplex mode. Its primary role is *security*: it transmits an artificial-noise (jamming) signal using beamformer $\mathbf{v}_J[n] \in \mathbb{C}^{N_J \times 1}$ that is designed to degrade the signal-to-interference-plus-noise ratio (SINR) at every eavesdropper while remaining as transparent as possible to the legitimate user, thereby increasing the achievable secrecy rate. In the proposed (theoretical) architecture, this UAV additionally receives the target-reflected sensing echoes – both the bistatic echo illuminated by the BS and the monostatic echo illuminated by itself – and combines them through selection combining; the associated self-interference-cancelled echo-reception chain is discussed as a modeling target in Section II-C.
- **Mobile ground user:** A single-antenna legitimate receiver located at $\mathbf{q}_U[n] \in \mathbb{R}^3$, which moves according to a random-walk mobility model and receives its information exclusively through the RIS-reflected path.
- **Vehicle sensing target:** A ground vehicle located at $\mathbf{q}_V[n] \in \mathbb{R}^3$, modeled as a point target with an aspect-angle-dependent radar cross section (RCS). The vehicle is the object of interest for the sensing sub-system and does not participate in the communication link.
- **HPPP-distributed eavesdroppers:** A field of passive, non-colluding eavesdroppers with locations

$$\Phi_E = \{\mathbf{q}_{E,e}\}_{e=1}^{N_E}, \qquad N_E \sim \mathrm{Poisson}(\lambda_E|\mathcal{A}|), \quad \mathbf{q}_{E,e} \sim \mathrm{Uniform}(\mathcal{A}), \tag{1}$$

  where λ_E is the spatial density of eavesdroppers and $\mathcal{A}$ is the bounded simulation area. Each eavesdropper attempts to overhear the RIS-reflected BS transmission intended for the user.

Fig. 1 summarizes the placement of the implemented entities. The two UAVs cooperate *asymmetrically*: the RIS-mounted UAV strengthens the legitimate BS–user path through passive (or learned) phase alignment, while the jammer UAV injects cooperative jamming toward the eavesdroppers and simultaneously supports the sensing function. Vehicle reflections and

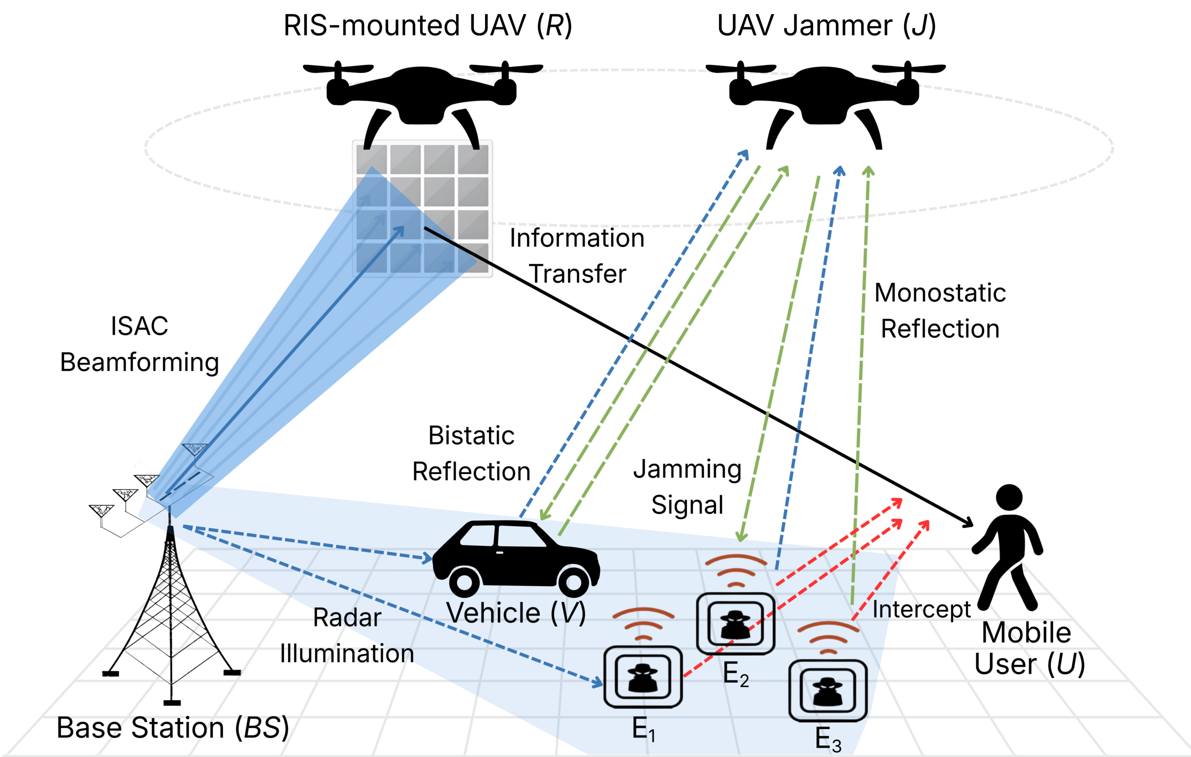


Fig. 1: Implemented ISAC-based UAV architecture: the RIS-mounted UAV relays the BS signal to the legitimate user, while the full-duplex jammer UAV protects the user by suppressing eavesdropper reception and participates in target sensing.

eavesdropper geometry are evaluated by dedicated sensing and secrecy modules described in Sections II-B and II-C. All wireless links use the generic distance-dependent path-loss and Rician small-scale fading model

$$g_{ab} = \beta_0 d_{ab}^{-\alpha} |h_{ab}|^2, \qquad d_{ab} = \|\mathbf{q}_a - \mathbf{q}_b\|, \tag{2}$$

$$h_{ab} = \sqrt{\frac{K}{K+1}}\, h_{\mathrm{LoS}} + \sqrt{\frac{1}{K+1}}\, h_{\mathrm{NLoS}}, \tag{3}$$

where β_0 is the reference channel gain at 1 m, α is the path-loss exponent, and K is the Rician factor ($K = 5$ in the representative implementation). This generic link model in (2)–(3) is instantiated separately for every BS–RIS, RIS–user, RIS–eavesdropper, jammer–user, jammer–eavesdropper, and sensing-related link introduced in the following two subsections.

B. Communication Model

1) Transmitted Signal and BS Beamforming: At time slot n, the BS transmits the beamformed communication signal

$$x_B[n] = \mathbf{w}_B[n]\, s[n], \tag{4}$$

where $s[n]$ is the unit-power information symbol, $\mathbb{E}\{|s[n]|^2\} = 1$, and $\mathbf{w}_B[n] \in \mathbb{C}^{M_B \times 1}$ is the BS beamforming vector, subject to the transmit-power constraint

$$P_B[n] = \|\mathbf{w}_B[n]\|_2^2 \le P_{B,\max}. \tag{5}$$

2) RIS-Assisted Channel: No Direct BS–User Link: Since the BS-to-user link is assumed unreliable/blocked, communication is carried exclusively through the RIS-mounted UAV. Three cascaded links are therefore required:

- $\mathbf{h}_{BR}[n] \in \mathbb{C}^{N_R \times M_B}$: the BS-to-RIS channel matrix,
- $\mathbf{h}_{RU}[n] \in \mathbb{C}^{N_R \times 1}$: the RIS-to-user channel vector,
- $\mathbf{h}_{RE,e}[n] \in \mathbb{C}^{N_R \times 1}$: the RIS-to-eavesdropper-e channel vector,

each generated according to the generic path-loss/Rician model of (2)–(3). The passive RIS applies the diagonal, unit-modulus phase-shift (reflection) matrix

$$\mathbf{\Phi}[n] = \mathrm{diag}\left(e^{j\phi_1[n]}, e^{j\phi_2[n]}, \ldots, e^{j\phi_{N_R}[n]}\right), \tag{6}$$

where $\phi_i[n] \in [-\pi, \pi]$ is the phase shift applied by the i-th RIS element and ideal unit-amplitude reflection is assumed.

a) Derivation of the effective cascaded channel $\mathbf{h}_{\text{eff}}$*.:* The BS signal first propagates to the i-th RIS element through $[\mathbf{h}_{BR}[n]]_i$, is phase-shifted by $e^{j\phi_i[n]}$, and is then re-radiated toward the user through $[\mathbf{h}_{RU}[n]]_i$. Summing the contribution of all N_R elements gives the scalar cascaded (effective) BS–RIS–user channel

$$h_{\text{eff},U}[n] = \sum_{i=1}^{N_R} [\mathbf{h}_{BR}[n]]_i \, e^{j\phi_i[n]} \, [\mathbf{h}_{RU}[n]]_i^*, \tag{7}$$

which, in compact matrix form, is written as

$$h_{\text{eff},U}[n] = \mathbf{h}_{RU}^H[n] \, \mathbf{\Phi}[n] \, \mathbf{h}_{BR}[n] \;\equiv\; \mathbf{h}_{BR}[n] \, \mathbf{\Phi}[n] \, \mathbf{h}_{RU}[n], \tag{8}$$

where the second, symmetric form is the notation used throughout this report, with $\mathbf{h}_{BR}$ and $\mathbf{h}_{RU}$ interpreted as the transmit- and receive-side RIS link vectors, respectively, and $(\cdot)^H$ denotes the conjugate transpose. When the BS uses $M_B > 1$ antennas, $\mathbf{h}_{BR}[n]$ is an $N_R \times M_B$ matrix and $h_{\text{eff},U}[n]$ in (8) becomes a $1 \times M_B$ effective row vector that is subsequently combined with the BS beamformer $\mathbf{w}_B[n]$. The identical construction applies to the e-th eavesdropper's leakage channel,

$$h_{\text{eff},E_e}[n] = \mathbf{h}_{RE,e}^H[n] \, \mathbf{\Phi}[n] \, \mathbf{h}_{BR}[n]. \tag{9}$$

The resulting beamformed channel powers at the user and at eavesdropper e are

$$G_U[n] = \eta_R \left| h_{\text{eff},U}[n] \, \mathbf{w}_B[n] \right|^2, \qquad G_{E,e}[n] = \eta_R \left| h_{\text{eff},E_e}[n] \, \mathbf{w}_B[n] \right|^2, \tag{10}$$

where $\eta_R \leq 1$ is the RIS reflection efficiency (ideal case $\eta_R = 1$).

3) Received Signals and Jammer Interference: The full-duplex jammer UAV contributes additional interference at every receiver through its own channels $\mathbf{h}_{JU}[n]$ (jammer-to-user) and $\mathbf{h}_{JE,e}[n]$ (jammer-to-eavesdropper-e), generated using the same generic link model. The signal received at the legitimate user is

$$y_U[n] = h_{\text{eff},U}[n] \, \mathbf{w}_B[n] \, s[n] + \mathbf{h}_{JU}[n] \, \mathbf{v}_J[n] \, z_J[n] + n_U[n], \tag{11}$$

and at eavesdropper e,

$$y_{E,e}[n] = h_{\text{eff},E_e}[n] \, \mathbf{w}_B[n] \, s[n] + \mathbf{h}_{JE,e}[n] \, \mathbf{v}_J[n] \, z_J[n] + n_{E,e}[n], \tag{12}$$

where $\mathbf{v}_J[n] \in \mathbb{C}^{N_J \times 1}$ is the jammer beamforming vector with power constraint $P_J[n] = \|\mathbf{v}_J[n]\|_2^2 \leq P_{J,\max}$, $z_J[n]$ is the zero-mean, unit-power artificial-noise symbol, and $n_U[n]$, $n_{E,e}[n] \sim \mathcal{CN}(0, \sigma^2)$ are additive thermal-noise terms. The corresponding jamming-channel gains are

$$J_U[n] = \left| \mathbf{h}_{JU}[n] \, \mathbf{v}_J[n] \right|^2, \qquad J_{E,e}[n] = \left| \mathbf{h}_{JE,e}[n] \, \mathbf{v}_J[n] \right|^2. \tag{13}$$

4) SINR, Capacity, and RIS-Assisted Secrecy Rate: Using (10) and (13), the user and worst-case eavesdropper SINR values are

$$\gamma_U[n] = \frac{G_U[n]}{P_J[n] \, J_U[n] + \sigma^2}, \qquad \gamma_{E,e}[n] = \frac{G_{E,e}[n]}{P_J[n] \, J_{E,e}[n] + \sigma^2}. \tag{14}$$

The instantaneous achievable rates of the legitimate user and of the strongest eavesdropper are

$$R_U[n] = \log_2(1 + \gamma_U[n]), \qquad R_{E,\max}[n] = \max_{e \in \{1,\ldots,N_E\}} \log_2(1 + \gamma_{E,e}[n]), \tag{15}$$

and the implemented per-slot secrecy rate follows the standard non-negative secrecy-capacity definition,

$$R_s[n] = \left[R_U[n] - R_{E,\max}[n] \right]^+, \qquad [x]^+ = \max(x, 0), \tag{16}$$

with aggregate secrecy rate over N time slots

$$R_{s,\text{total}} = \sum_{n=1}^{N} R_s[n]. \tag{17}$$

The full-duplex jammer therefore protects the user by simultaneously (i) suppressing $R_{E,\max}[n]$ through $J_{E,e}[n]$ in (14), and (ii) being designed, through the choice of $\mathbf{v}_J[n]$ (e.g., nullspace-oriented beamforming toward the user direction), to keep $J_U[n]$ small so that $R_U[n]$ is largely preserved. Equation (16) is the primary secrecy objective used by the optimization and MADRL evaluators of this system.

C. Sensing Model

1) Sensing Geometry: Monostatic and Bistatic Paths: Sensing is performed against the ground vehicle target located at $\mathbf{q}_V[n]$. Two complementary sensing geometries are considered:

- **Monostatic sensing:** the RIS-mounted UAV, acting as a co-located transmit/receive uniform linear array (ULA), illuminates the vehicle and receives its own reflected echo.
- **Bistatic sensing:** the BS illuminates the vehicle, and the target-reflected echo is received at the full-duplex jammer UAV, which is spatially separated from the BS.

Both paths use the same RCS-based target reflection model, and the two resulting SNRs are subsequently combined through selection combining, as described below.

2) Target Reflection and Radar Cross Section: Each vehicle is treated as a point target whose reflectivity depends on its type and aspect angle ψ through the RCS

$$\sigma_{\text{rcs}}(\text{type}, \psi) = 10^{\,\text{RCS}_{\text{dB}}(\text{type})/10}\ \max\bigl(\cos^2\psi,\, 0.01\bigr)\,. \tag{18}$$

For the monostatic path (sensing platform $P \in \{R\}$ transmitting to, and receiving from, the vehicle V), the scalar round-trip reflection channel is

$$h_{\text{mono},k} = h_{P,V_k}\,\sqrt{\sigma_{\text{rcs},k}}\,h_{V_k,P}, \tag{19}$$

and for the bistatic path (BS transmitting, jammer UAV receiving),

$$h_{\text{bi},k} = h_{B,V_k}\,\sqrt{\sigma_{\text{rcs},k}}\,h_{V_k,J}, \tag{20}$$

where h_{P,V_k}, h_{B,V_k}, $h_{V_k,P}$, and $h_{V_k,J}$ are individually generated using the generic path-loss/Rician link model of (2)–(3), with k indexing the vehicle target.

3) Received Sensing Signal and Sensing SNR: For a transmitted sensing signal of power P_s, the scalar echo received at the sensing receiver (monostatic or bistatic) is

$$y_{\text{sens},k} = \sqrt{P_s}\,h_{(\cdot),k}\,s[n] + n_{\text{sens}}[n], \qquad n_{\text{sens}}[n] \sim \mathcal{CN}(0,\sigma_n^2), \tag{21}$$

where $h_{(\cdot),k}$ denotes either $h_{\text{mono},k}$ or $h_{\text{bi},k}$. The received sensing power and the sensing SNR follow directly as

$$G_{\text{sens},k} = \bigl|h_{(\cdot),k}\bigr|^2, \qquad \text{SNR}_{(\cdot),k} = \frac{P_s\,G_{\text{sens},k}}{\sigma_n^2}. \tag{22}$$

4) Selection Combining Between Monostatic and Bistatic Sensing: Since the monostatic and bistatic paths illuminate the same vehicle target from different geometries, they generally experience independent fading and path-loss conditions. To exploit this diversity, the jammer UAV applies selection combining (SC) across the two branches, retaining the branch with the larger instantaneous sensing SNR:

$$\text{SNR}_{\text{SC}} = \max\left(\text{SNR}_{\text{mono}},\, \text{SNR}_{\text{bi}}\right). \tag{23}$$

Equation (23) is used as the effective post-combining sensing SNR that feeds subsequent estimation and detection processing, so that the sensing sub-system is never worse off than the better of its two individual sensing links.

5) Array Model and Composite Sensing Channel: The sensing platform is modeled as an N-element ULA with steering vector

$$\mathbf{a}(\theta) = \Bigl[1,\ e^{-j\frac{2\pi d}{\lambda}\sin\theta},\ \ldots,\ e^{-j\frac{2\pi(N-1)d}{\lambda}\sin\theta}\Bigr]^T, \tag{24}$$

where d is the inter-element spacing and λ is the carrier wavelength. For the k-th target at angle θ_k, the (monostatic) target response matrix and the composite multi-target sensing channel are

$$\mathbf{A}_k = \mathbf{a}(\theta_k)\,\mathbf{a}^H(\theta_k), \qquad \mathbf{H}_{\text{sense}} = \sum_k \alpha_k\,\mathbf{A}_k, \tag{25}$$

where α_k is the complex reflection coefficient of target k (incorporating $\sigma_{\text{rcs},k}$ and path loss). Random unit-norm pilot columns $\mathbf{x}_l \sim \mathcal{CN}(\mathbf{0},\mathbf{I})$, normalized as $\mathbf{X}[:,l] = \mathbf{x}_l/\|\mathbf{x}_l\|_2$, are transmitted over L pilot symbols, and the received echo matrix is

$$\mathbf{Y} = \mathbf{H}_{\text{sense}}\mathbf{X} + \mathbf{N}, \qquad [\mathbf{N}]_{ij} \sim \mathcal{CN}(0,\sigma^2). \tag{26}$$

6) Angle Estimation: Fisher Information and Cramer–Rao Bound: Target angle is first estimated geometrically as

$$\theta_k = \text{atan2}(y_k - y_{\text{sensor}},\ x_k - x_{\text{sensor}}), \tag{27}$$

and its estimation accuracy is characterized through the Fisher information matrix (FIM) $\mathbf{F}_{\boldsymbol{\theta}}$ for the target-angle vector $\boldsymbol{\theta} = [\theta_1,\ldots,\theta_K]^T$. Using the steering-vector derivative

$$\frac{\partial a_n(\theta)}{\partial\theta} = -j\frac{2\pi n d}{\lambda}\cos\theta\; a_n(\theta), \tag{28}$$

the target-response and channel derivatives are

$$\frac{\partial\mathbf{A}_k}{\partial\theta_k} = \dot{\mathbf{a}}_k\mathbf{a}_k^H + \mathbf{a}_k\dot{\mathbf{a}}_k^H, \qquad \frac{\partial\mathbf{H}_{\text{sense}}}{\partial\theta_k} = \alpha_k\frac{\partial\mathbf{A}_k}{\partial\theta_k}, \tag{29}$$

and the FIM entries are

$$[\mathbf{F}_{\boldsymbol{\theta}}]_{ij} = \frac{2}{\sigma^2}\,\text{Re}\left\{\text{tr}\left(\left(\frac{\partial\mathbf{H}_{\text{sense}}}{\partial\theta_i}\right)^H \frac{\partial\mathbf{H}_{\text{sense}}}{\partial\theta_j}\,\mathbf{X}\mathbf{X}^H\right)\right\}. \tag{30}$$

The Cramer–Rao bound matrix, per-target variance bound, RMSE bound, and scalar sensing-accuracy metric are then

$$\text{CRB} = \mathbf{F}_{\boldsymbol{\theta}}^{-1}, \qquad \text{var}_k = [\text{CRB}]_{k,k}, \qquad \text{rmse}_k = \sqrt{\text{var}_k}, \qquad \text{CRB}_{\text{trace}} = \text{tr}(\text{CRB}), \tag{31}$$

where $\text{CRB}_{\text{trace}}$ lower-bounds the total mean-squared angle-estimation error and is used as the primary sensing-accuracy scalar throughout the optimization and MADRL objectives.

7) Target Detection: Energy Detector and GLRT: Target presence is formulated as the binary hypothesis test

$$\mathcal{H}_0 : \; \mathbf{Y} = \mathbf{N}, \qquad \mathcal{H}_1 : \; \mathbf{Y} = \mathbf{H}_{\text{sense}}\mathbf{X} + \mathbf{N}. \tag{32}$$

The energy-detector test statistic is

$$T_E(\mathbf{Y}) = \|\mathbf{Y}\|_F^2, \tag{33}$$

and the detection and false-alarm probabilities, together with their Monte Carlo estimators over M trials, are

$$P_D = \Pr\{T(\mathbf{Y}) > \eta \mid \mathcal{H}_1\}, \qquad P_{FA} = \Pr\{T(\mathbf{Y}) > \eta \mid \mathcal{H}_0\}, \tag{34}$$

$$\hat{P}_D = \frac{1}{M}\sum_{m=1}^{M} \nVdash\big[T(\mathbf{Y}_m^1) > \eta\big]\,, \qquad \hat{P}_{FA} = \frac{1}{M}\sum_{m=1}^{M} \nVdash\big[T(\mathbf{Y}_m^0) > \eta\big]\,. \tag{35}$$

Under $\mathcal{H}_0$, the energy-detector statistic follows a Gamma distribution, allowing an analytic Neyman–Pearson threshold to be set for a target false-alarm rate:

$$T_E \mid \mathcal{H}_0 \sim \text{Gamma}\big(N_r L,\, \sigma^2\big)\,, \qquad \eta = F_{\text{Gamma}}^{-1}\big(1 - P_{FA};\, N_r L,\, \sigma^2\big)\,. \tag{36}$$

As an alternative, higher-resolution statistic, the generalized-likelihood-ratio-test (GLRT)-style detector is

$$\mathbf{P}_{\mathbf{X}} = \mathbf{X}^H\big(\mathbf{X}\mathbf{X}^H + \epsilon\mathbf{I}\big)^{-1}\mathbf{X}, \qquad T_{\text{GLRT}}(\mathbf{Y}) = \frac{\|\mathbf{Y}\mathbf{P}_{\mathbf{X}}\|_F^2}{\|\mathbf{Y} - \mathbf{Y}\mathbf{P}_{\mathbf{X}}\|_F^2}. \tag{37}$$

The detection probability P_D from (34)–(35), evaluated on either T_E or T_{GLRT} after the selection-combining stage of (23), and the CRB-based accuracy metric of (31) together constitute the two complementary sensing-quality indicators used throughout the remainder of this report.

III. Proposed Methodology

The proposed methodology integrates the communication, sensing, and optimization components described in Section II into a complete operational workflow. The system operates over discrete time slots indexed by $n = 1, \dots, N$. At each slot, the BS beamforms an information-bearing signal toward the RIS-mounted UAV, which applies a passive phase-shift matrix and reflects the signal toward the ground user. Simultaneously, the full-duplex jammer UAV transmits artificial interference designed to suppress eavesdropper reception while minimally affecting the legitimate user. The RIS-UAV trajectory, BS beamforming, jammer beamforming, and RIS phase configuration constitute the set of controllable degrees of freedom that determine both the achievable secrecy rate and the sensing accuracy. These four decision spaces are collectively optimized through a multi-agent deep reinforcement learning framework.

A. Communication Pipeline

The communication pipeline computes the secrecy rate that the cooperative BS–RIS–jammer architecture achieves against spatially distributed eavesdroppers. At each slot n, the BS transmits the beamformed signal $x_B[n] = \mathbf{w}_B[n]s[n]$ as defined in (4), subject to the per-slot power constraint $\|\mathbf{w}_B[n]\|_2^2 \le P_{B,\max}$ from (5). The BS beamforming vector $\mathbf{w}_B[n]$ has dimensionality M_B, the number of BS antennas.

The beamformed signal propagates through the BS–RIS channel $\mathbf{h}_{BR}[n]$ and is phase-shifted elementwise by the RIS reflection matrix $\mathbf{\Phi}[n]$ of (6). For the legitimate user, the effective cascaded channel is $h_{\text{eff},U}[n] = \mathbf{h}_{RU}^H[n]\mathbf{\Phi}[n]\mathbf{h}_{BR}[n]$ from (8), and for each eavesdropper e, the leakage channel is $h_{\text{eff},E_e}[n] = \mathbf{h}_{RE,e}^H[n]\mathbf{\Phi}[n]\mathbf{h}_{BR}[n]$ from (9). The beamformed channel power at the user and at each eavesdropper follow (10).

The full-duplex jammer UAV simultaneously transmits an artificial-noise signal using beamformer $\mathbf{v}_J[n] \in \mathbb{C}^{N_J \times 1}$ with power constraint $\|\mathbf{v}_J[n]\|_2^2 \le P_{J,\max}$. The jammer-to-user and jammer-to-eavesdropper gains $J_U[n]$ and $J_{E,e}[n]$ are given by (13). The jammer beamformer $\mathbf{v}_J[n]$ is the third controlled variable; it can be configured in various heuristic modes—including isotropic, nullspace-oriented, and mixed strategies—or, in the learning framework, produced directly by the jammer agent.

The user SINR $\gamma_U[n]$ and per-eavesdropper SINR $\gamma_{E,e}[n]$ follow (14), with thermal noise variance σ^2. The instantaneous achievable rates $R_U[n]$ and $R_{E,\max}[n]$ are computed via (15), leading to the per-slot secrecy rate $R_s[n] = [R_U[n] - R_{E,\max}[n]]^+$ from (16). The aggregate secrecy objective over the N-slot horizon is $R_{s,\text{total}} = \sum_{n=1}^{N} R_s[n]$ from (17).

The number and positions of eavesdroppers are not fixed but are drawn at each environment reset from a homogeneous Poisson point process with density λ_E over the bounded simulation area $\mathcal{A}$, as described in (1). The active eavesdropper count is capped at a fixed maximum $N_{E,\max}$ for tractable tensor representation; any inactive slots are placed at a far-away coordinate such that their channel contribution becomes negligible.

The mobile ground user moves according to a random-walk mobility model. At each time step, the user position $\mathbf{q}_U[n]$ is updated by a random displacement drawn from a zero-mean Gaussian distribution with configurable step variance, and the position is clipped to remain within the simulation workspace. This mobility pattern introduces dynamic channel conditions that require the learning agents to adapt their policies continuously rather than converging to a static geometry.

B. Sensing Pipeline

The sensing pipeline performs target angle estimation and detection using the same RIS-mounted UAV that serves the communication link, which couples the two objectives through the shared trajectory decision $\mathbf{q}_R[n]$. The sensing target is a ground vehicle modeled as a point reflector with radar cross section σ_{rcs} that depends on vehicle type and aspect angle ψ as given in (18). The RIS-mounted UAV serves as the primary sensing platform: its uniform linear array (ULA) with N elements, steering vector $\mathbf{a}(\theta)$ from (24), and position $\mathbf{q}_R[n]$ collectively determine the target geometry and the resulting sensing channel.

For target angle estimation, the composite sensing channel matrix $\mathbf{H}_{\text{sense}}$ is constructed from per-target response matrices $\mathbf{A}_k = \mathbf{a}(\theta_k)\mathbf{a}^H(\theta_k)$ weighted by complex reflection coefficients α_k, as in (25). Unit-norm pilot symbols $\mathbf{X}$ are transmitted, and the received echo matrix $\mathbf{Y} = \mathbf{H}_{\text{sense}}\mathbf{X} + \mathbf{N}$ follows (26). The Fisher information matrix $\mathbf{F}_{\boldsymbol{\theta}}$ for the target-angle vector is built from channel derivatives with respect to each target angle using (30), and the Cramer–Rao bound matrix is $\text{CRB} = \mathbf{F}_{\boldsymbol{\theta}}^{-1}$ from (31). The scalar sensing-accuracy metric employed in the optimization and learning objectives is the CRB trace, $\text{CRB}_{\text{trace}} = \text{tr}(\text{CRB})$, which lower-bounds the total mean-squared angle-estimation error across all targets.

To convert the CRB trace into a sensing utility suitable for inclusion in a scalar objective, the proposed framework applies the logarithmic transformation

$$U_{\text{sens}}[n] = -\log_{10}\!\big(\max(\text{CRB}_{\text{trace}}[n], \epsilon)\big), \tag{38}$$

where ϵ is a small constant preventing numerical divergence. The aggregate sensing utility over N slots is $U_{\text{sens,total}} = \sum_{n=1}^{N} U_{\text{sens}}[n]$.

Target detection is formulated as the binary hypothesis test of (32) and evaluated using two complementary detectors. The energy detector uses the Frobenius-norm statistic $T_E(\mathbf{Y}) = \|\mathbf{Y}\|_F^2$ from (33), and the generalized-likelihood-ratio-test (GLRT) detector uses the projection-based statistic $T_{\text{GLRT}}(\mathbf{Y})$ from (37). Detection and false-alarm probabilities are estimated via Monte Carlo simulation as in (35). Under the null hypothesis, the energy detector follows a Gamma distribution, enabling an analytic Neyman–Pearson threshold for a target false-alarm rate, as in (36). The detection probability P_D is computed within the sensing modules but does not directly appear in the primary secrecy–CRB optimization objective of the main ISAC pipeline. A separate sensing-only multi-agent study—described in Section III-D—evaluates P_D as an explicit reward component.

C. Joint Optimization Framework

The communication and sensing pipelines of Sections III-A and III-B each depend on the same four sets of controllable parameters—the BS beamforming vectors $\{\mathbf{w}_B[n]\}$, the jammer beamforming vectors $\{\mathbf{v}_J[n]\}$, the RIS-UAV trajectory positions $\{\mathbf{q}_R[n]\}$, and the RIS phase shifts $\{\phi_i[n]\}$—but they pursue distinct and partially conflicting objectives. Maximizing only the secrecy rate tends to favour configurations that weaken the sensing geometry, while maximizing only the sensing accuracy neglects the security of the communication link. A joint formulation is therefore required to capture this trade-off explicitly within a single optimization framework.

The overall system behavior is governed by a joint optimization problem that balances communication secrecy and sensing accuracy under physical feasibility constraints. The optimization variables are the four sets of controllable parameters: the BS beamforming vectors $\{\mathbf{w}_B[n]\}_{n=1}^{N}$, the jammer beamforming vectors $\{\mathbf{v}_J[n]\}_{n=1}^{N}$, the RIS-UAV trajectory positions $\{\mathbf{q}_R[n]\}_{n=1}^{N}$, and the RIS phase shifts $\{\phi_i[n]\}_{i=1}^{N_R}$ comprising $\mathbf{\Phi}[n]$.

The scalar objective combines the secrecy and sensing metrics into a single weighted function:

$$F(\alpha) = \alpha\,\frac{R_{s,\text{total}}}{R_{S,\text{REF}}} + (1-\alpha)\,\frac{U_{\text{sens,total}}}{U_{\text{REF}}}, \tag{39}$$

where $\alpha \in [0,1]$ is a trade-off parameter that controls the relative emphasis on secrecy versus sensing accuracy. The reference values $R_{S,\text{REF}}$ and U_{REF} normalize the two components to comparable magnitudes. The secrecy component $R_{s,\text{total}}$ is computed through the communication pipeline described in Section III-A, and the sensing component $U_{\text{sens,total}}$ follows the CRB-based utility of Section III-B.

The objective is subject to several constraint sets. The BS and jammer transmit powers are bounded per slot by $P_{B,\max}$ and $P_{J,\max}$, respectively, as in (5). The RIS reflection matrix $\mathbf{\Phi}[n]$ must satisfy the ideal unit-modulus condition $|e^{j\phi_i[n]}| = 1$ from (6). The RIS-UAV trajectory must remain within a three-dimensional workspace $\mathbf{q}_{\min} \preceq \mathbf{q}_R[n] \preceq \mathbf{q}_{\max}$ and must satisfy a per-slot speed constraint $\|\mathbf{q}_R[n] - \mathbf{q}_R[n-1]\|_2/\Delta t \le v_{\max}$, where Δt is the slot duration and $v_{\max}$ is the maximum UAV speed.

Rather than solving (39) directly through conventional nonlinear programming—which is challenging due to its non-convexity and the coupling of discrete RIS phases with continuous beamforming and trajectory variables—the implemented system adopts a multi-agent deep reinforcement learning approach. The objective $F(\alpha)$ serves as the core of the shared reward signal, and the four decision spaces are assigned to independent learning agents that cooperatively maximize the reward through interaction with the environment. Legacy successive convex approximation (SCA) and block coordinate descent (BCD) methods serve as supporting baselines but are superseded by the MADRL framework for the primary ISAC workflow.

D. Multi-Agent Deep Reinforcement Learning Control

The MADRL framework defines four cooperative agents whose collective behavior determines the system state and the resulting joint objective value. Each agent observes a task-specific subset of the environment state, outputs a continuous action vector, and contributes to a shared global reward.

1) Agent Architecture: **BS Beamforming Agent.** This agent determines the BS beamforming vector $\mathbf{w}_B[n]$ that directs the information-bearing signal toward the RIS-mounted UAV. The agent observes the effective cascaded user and eavesdropper channels together with the current secrecy and sensing metrics, and outputs a continuous action that is projected onto the feasible power ball $\|\mathbf{w}_B[n]\|_2^2 \leq P_{B,\max}$. The action can be represented as a direct complex-coefficient vector, a unit-direction vector scaled to full power, or a joint power-fraction and direction output, each providing a different exploration characteristic during learning.

UAV Trajectory Agent. This agent controls the RIS-UAV position $\mathbf{q}_R[n]$, which determines both the communication channel quality and the sensing geometry. Its observation includes the positions of the vehicles, user, and eavesdroppers, together with the previous trajectory waypoint and the current performance metrics. The action is a position increment $\Delta\mathbf{q}_R[n]$, scaled by the maximum permissible step $v_{\max}\Delta t$, clipped to the three-dimensional workspace $[\mathbf{q}_{\min}, \mathbf{q}_{\max}]$, and constrained by the per-slot speed limit $\|\Delta\mathbf{q}_R[n]\|_2 \leq v_{\max}\Delta t$.

Jammer Beamforming Agent. This agent controls the jammer beamforming vector $\mathbf{v}_J[n]$ that radiates artificial interference toward the eavesdroppers. The agent observes aggregate eavesdropper SINR, jammer power utilization, and the current performance metrics, then outputs a continuous action that is projected onto the feasible set $\|\mathbf{v}_J[n]\|_2^2 \leq P_{J,\max}$.

RIS Phase Agent. This agent controls the individual phase shifts $\phi_i[n]$ of the RIS reflection matrix $\mathbf{\Phi}[n]$, thereby shaping the effective cascaded channel without requiring additional radio-frequency hardware. Its observation includes the previous phase configuration and a heuristic alignment hint computed from the cascaded channel angles. The action outputs phase values in $[-\pi, \pi]$ per RIS element, which are applied directly to $\mathbf{\Phi}[n]$ in (6).

2) Reward: All four agents share the same scalar reward, which is based on the joint objective of (39) augmented with penalty terms for constraint violations and secrecy outage:

$$r = \alpha \, \frac{R_{s,\text{total}}}{R_{S,\text{REF}}} + (1-\alpha) \, \frac{U_{\text{sens,total}}}{U_{\text{REF}}} - \lambda_c \, v_c - \lambda_o \, v_o, \tag{40}$$

where v_c measures the total violation of the physical constraints (BS power, jammer power, RIS unit modulus, and UAV speed and workspace limits), v_o captures the secrecy-rate shortfall when $R_{s,\text{total}}$ falls below a minimum threshold, and the coefficients λ_c and λ_o control the severity of each penalty. The coefficient λ_c therefore encourages the agents to respect the power, RIS, and trajectory limits, while λ_o discourages episodes in which the secrecy rate is unacceptably low. Additional reward normalisation strategies—such as running z-score normalization of the secrecy and sensing components—can be employed to stabilize training when the metric magnitudes are poorly calibrated.

3) Training Procedure: The environment is episodic, with each episode spanning N time slots. At the start of each episode, the UAV trajectory is initialised along a straight-line path, the BS and jammer beamformers are initialised to feasible values, the RIS phases are initialised to zero, the user is placed at a starting position, and a fresh set of eavesdroppers is sampled from the HPPP. At each step within the episode, every agent produces a continuous action from its local observation, the environment projects each action onto the corresponding feasible set, the user advances according to the random-walk mobility model, and the secrecy rate, CRB-derived sensing utility, constraint violations, and reward are recomputed.

The primary training algorithms are multi-agent proximal policy optimization (MAPPO) and multi-agent twin-delayed deep deterministic policy gradient (MATD3PG). MAPPO is an on-policy actor–critic method: each agent maintains a policy network and a value function; the policy is updated by maximizing a clipped surrogate objective using advantage estimates computed from trajectory batches, and the value function is updated by minimizing the mean-squared temporal-difference error. MATD3PG extends the twin-delayed DDPG framework to the multi-agent setting, employing twin Q-networks to mitigate overestimation bias, delayed policy updates for stability, and target policy smoothing for robust value estimation.

Convergence is assessed by monitoring the per-agent episode rewards, the aggregate secrecy rate $R_{s,\text{total}}$, the aggregate sensing utility $U_{\text{sens,total}}$, and the constraint-violation magnitude over successive training episodes.

In addition to the primary ISAC MADRL framework, a sensing-only study evaluates cooperative sensing policies using MAPPO, MATD3PG, MADDPG (multi-agent deep deterministic policy gradient), and a Random Walk baseline. In that study, each agent controls a sensing UAV moving in two dimensions, observes its own CRB utility and detection probability, and receives a team reward that explicitly balances P_D and CRB utility. The Random Walk baseline displaces each sensing UAV by a random velocity at each step, providing a non-learning reference. This sensing-only study complements the main ISAC pipeline by demonstrating detection-probability optimization in isolation, while the primary ISAC system remains focused on the joint secrecy–CRB objective of (39).

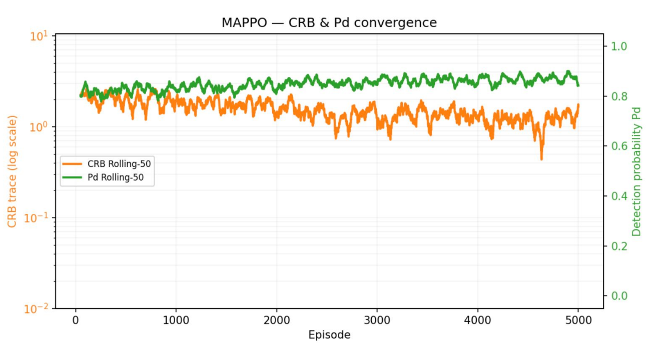
At the end of each time slot, the complete methodology has executed the following sequence: the BS beamforms the communication signal toward the RIS, which reflects it toward the ground user; the jammer UAV radiates artificial interference toward the eavesdroppers; sensing measurements are collected and processed to obtain the CRB-based utility; the secrecy rate and sensing utility are combined into the weighted objective of (39); the shared reward is evaluated according to (40); and the four agents update their policies based on the observed outcome. This closed-loop interaction repeats over successive episodes, enabling the agents to learn cooperative control strategies that jointly optimize the secrecy–sensing trade-off without requiring explicit knowledge of the underlying channel or geometry dynamics.

IV. Results and Discussion

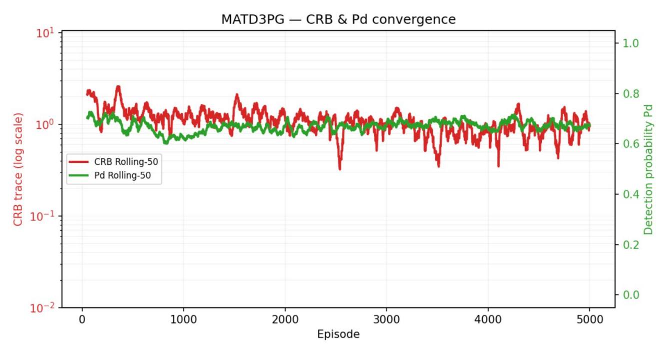
This section reports the training behaviour of the sensing-only multi-agent study introduced in Section III-D, in which MAPPO, MATD3PG, MADDPG, and a non-learning Random-Walk baseline are each trained for 5000 episodes on the cooperative two-UAV sensing task, with per-episode CRB trace and detection probability P_D tracked throughout training. All curves are smoothed with a rolling window of 50 episodes to make the underlying trend visible against the per-episode geometry-randomization noise, since vehicle-target and UAV start positions are re-sampled at every episode reset.

A. Per-Algorithm Convergence

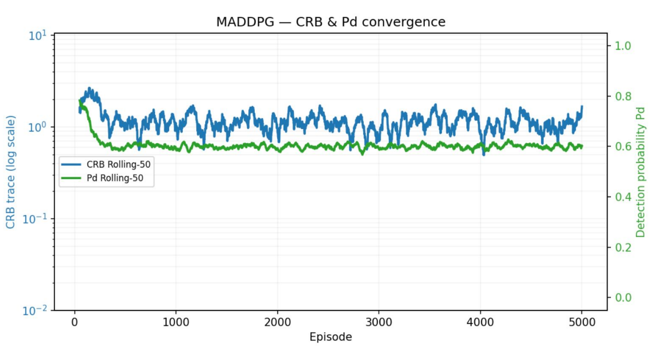
Figs. 2(a)–2(d) show the dual-axis CRB (left, log scale) and P_D (right) rolling-50 curves for MAPPO, MATD3PG, MADDPG, and the Random-Walk baseline, respectively.



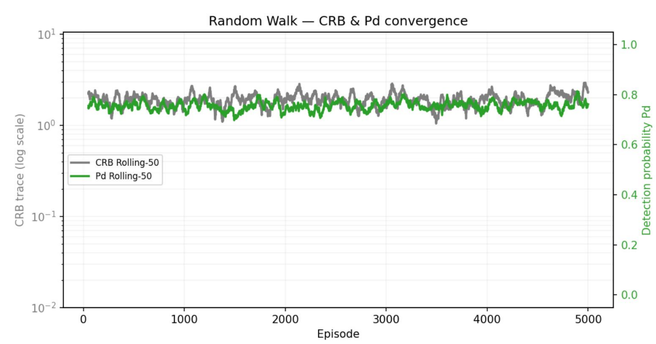
(a) MAPPO



(b) MATD3PG



(c) MADDPG

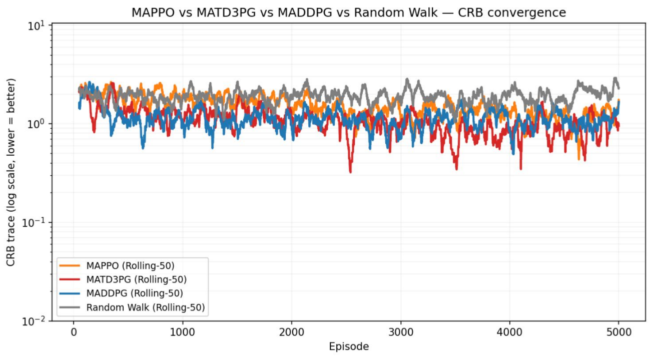


(d) Random Walk

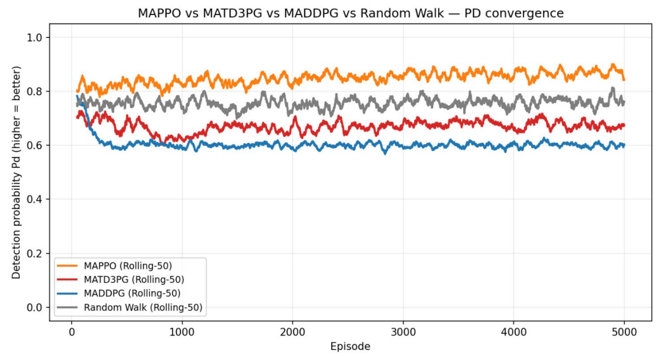
Fig. 2: Training convergence of the four evaluated algorithms. Each subplot shows the rolling-50 CRB trace (left axis, logarithmic scale) and detection probability P_D (right axis) over 5000 training episodes.

MAPPO (Fig. 2a) shows the clearest learning signal: P_D rises from approximately 0.82 early in training to a stable band around 0.85–0.88 by episode 5000, while the CRB trace trends downward, from roughly 2 to a noisy band between 1 and 1.5. MATD3PG (Fig. 2b) exhibits a different pattern: P_D initially declines from about 0.72 to a floor near 0.62–0.65 before recovering modestly to about 0.65–0.70, while its CRB trace is the noisiest of the four, with occasional deep troughs below 0.4 indicating episodes of especially favourable sensing geometry, but without a clear sustained downward trend. MADDPG (Fig. 2c) converges quickly – within roughly 300 episodes – to a flat, low-variance P_D plateau near 0.60, the lowest steady-state P_D of the three learned methods, while its CRB trace remains broadly comparable to MAPPO's. The Random-Walk baseline (Fig. 2d), as expected for a non-learning policy, shows no directional trend in either metric: P_D fluctuates around 0.75–0.80 and the CRB trace around 1.5–2.5 throughout, providing a stationary reference against which the three learned methods can be judged.

B. Comparative Analysis



(a) CRB convergence comparison



(b) Detection probability comparison

Fig. 3: Comparison of the four algorithms. (a) CRB convergence comparison and (b) detection probability (P_D) convergence comparison over 5000 training episodes.

Overlaying all four methods (Figs. 3a–3b) makes two observations clear. First, on the CRB metric (Fig. 3a), all three learned methods track a broadly similar band that sits modestly below the Random-Walk baseline for most of training, with MATD3PG reaching the lowest instantaneous troughs but MADDPG offering the most consistent, lowest-variance improvement; none of the three achieves a large, clean separation from the random baseline, suggesting that CRB in this cooperative two-agent formulation is comparatively easy to improve on only marginally through geometry alone and leaves limited additional headroom over undirected motion within the bounded scene. Second, and more strikingly, on the detection-probability metric (Fig. 3b) the three learned methods do not uniformly outperform the non-learning baseline: only MAPPO opens a clear, sustained margin above Random Walk (about 0.85–0.88 vs. 0.75–0.80), while both MATD3PG and MADDPG converge to P_D values *below* the random baseline. This is consistent with a known pathology of independent off-policy multi-agent learners under a shared team reward: with no explicit per-agent credit assignment, an off-policy critic can lock onto a low-exploration, low-variance local optimum early in training (MADDPG plateaus within roughly 300 episodes) that is stable but suboptimal for the coupled P_D objective, whereas MAPPO's on-policy updates and entropy regularization sustain broader exploration of the sensing geometry for longer, allowing it to continue improving P_D well past episode 1000. This asymmetry between the CRB and P_D trends – a meaningful separation on P_D but not on CRB – also indicates that the two sensing objectives respond differently to trajectory optimization in this scenario, and that reward-shaping or centralized-critic extensions to MATD3PG/MADDPG (Section III-D) are a natural next step for closing this gap.

V. Conclusion

This report presented the implementation of an ISAC-based UAV framework that integrates secure wireless communication, cooperative sensing, and multi-agent reinforcement learning within a unified simulation environment. The implemented architecture consists of a multi-antenna BS, a passive RIS-mounted UAV, a full-duplex cooperative jammer UAV, a mobile user, a vehicle sensing target, and HPPP-distributed eavesdroppers. The communication model incorporates RIS-assisted beamforming, secrecy-rate evaluation, and cooperative jamming, while the sensing framework implements vehicle reflection modeling, Fisher Information Matrix (FIM)-based Cramér–Rao Bound (CRB) analysis, energy detection, generalized likelihood ratio testing (GLRT), and detection probability evaluation.

The optimization framework combines communication and sensing objectives through a weighted secrecy–sensing formulation and is complemented by a four-agent MADRL framework that jointly optimizes BS beamforming, UAV trajectory, jammer beamforming, and RIS phase configuration. Comparative experiments using MAPPO, MATD3PG, MADDPG, and a Random Walk baseline demonstrate stable learning behaviour and validate the effectiveness of the proposed framework through representative convergence characteristics and sensing performance metrics. The implementation successfully validates the major mathematical models of the proposed ISAC system at the communication, sensing, optimization, and learning levels.

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